

Humean Analyses of Law and Chance

PHIL UA-78 Metaphysics

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Humean analyses of lawhood

An *analysis* of lawhood/nomic necessity or chance would be a claim of the form:

- ▶ For it to be nomically necessary [a law] that p is for it to be the case that ...
- ▶ For the objective chance of p (at t) to be x is for it to be the case that

A **Humean** analysis would be one where the right-hand-side doesn't contain expressions like 'law', 'chance', 'cause', 'produce', 'explain', 'would',

Anti-Humeans are, minimally, opposed to all Humean analyses.

The Naïve Regularity Theory

Carroll kicks off his discussion with the following analysis of *lawhood*:

The Naïve Regularity Theory

It is a law that all F s are G s if and only if (i) all F s are G s, and (ii) the generalization that all F s are G s is law-like; that is, (a) it is not necessary that all F s are G s, and (b) the generalization is unrestricted (it involves only non-local, empirical predicates apart from logical connectives and quantifiers).

I guess the view is that whenever it's a law that p , there are properties F and G such that p is the state of affairs that F s are G s.

The Naïve Regularity Theory

Predicates are pieces of language—this is confirmed when Carroll later explains ‘non-local’ by saying ‘local predicates include names of individual times, places, or objects’. But I don’t think we want ‘law’ to be a predicate of *sentences*! Rather, I suggest the following amendments:

- ▶ Replace ‘predicate’ with ‘property’.
- ▶ Understand ‘non-local’ as ‘qualitative’.
- ▶ Forget about ‘empirical’, which doesn’t seem to play any important role in Carroll’s discussion.
- ▶ The part about ‘not [metaphysically] necessary’ is also quite unimportant!

Carroll's objections to the Naïve Regularity Theory

- (i) Being a unicorn is qualitative (non-local) and uninstantiated, but it is not a law that all unicorns are F for any qualitative F .
- (ii) It entails that it's a law that all unicorns-or-ravens are black.
 - ▶ Note that the naïve regularity theorist is committed to its being a law that *all things are non-unicorns*, which is equivalent to *there are no unicorns*. *All unicorns are white* and *all unicorns are black* both follow from this. Carroll's argument about counterfactuals is a blunder.
- (iii) It entails that if it's true that all gold spheres are less than 10 feet in diameter, it's a law.
 - ▶ This is the real problem. It's even worse if we consider super-specific qualitative properties that give complete descriptions of someone's life. . .

The restriction in the NRT that requires laws to be states of affairs of the form *All Fs are Gs* does much less work than you might think. Assuming it's metaphysically necessary that there's at least one thing, any state of affairs p is necessarily equivalent to *all things are such-that- p* .

So if we don't believe in, or don't care about, differences between metaphysically necessarily equivalent states of affairs, the NRT is pretty much equivalent to the claim that *being a law is being a qualitative, metaphysically contingent truth*.

A more general format

Here's a general schema:

Lawhood = Lawlike Truth

To be a law of nature is to be a truth that is **lawlike**.

- Assume that if p is lawlike it is metaphysically necessarily lawlike, and if p is not lawlike it is metaphysically necessarily non-lawlike.

We recover the NRT if we flesh out 'lawlike' as something like *qualitative and metaphysically contingent*. But this lets in way too much.

But one can keep the general schema while having a different theory of lawlikeness. One prima facie promising idea is that being lawlike requires being (not only qualitative, but) *sufficiently simple*.

Standard objections to the 'lawlike truth' theory

- (i) It is not nomically necessary, though it may be true, that all gold spheres are less than a mile in diameter; but it is nomically necessary that all uranium spheres are less than a mile in diameter.
- (ii) It is not in fact nomically necessary that all gold spheres are less than a mile in diameter, but it is metaphysically possible for this to be nomically necessary.

Lawhood versus nomic necessity

The Lawlike Truth theory is hopeless as a theory of *nomic necessity*. Whenever p is nomically necessary, $p \vee q$ is nomically necessary too; but even if p is qualitative and extremely simple, $p \vee q$ might be non-qualitative and horrendously complicated.

If we're primarily interested in analysing nomic necessity, we can (i) treat Lawlike Truth as an analysis of "basic law", and (ii) analyze 'nomically necessary' as 'entailed by the basic laws'.

We can say that neither *no gold spheres are $>10ft$* nor *no uranium spheres are $>10ft$* is lawlike (too complex I guess?) But there are some lawlike truths that entail the latter. And there are some lawlike falsehoods which could be true and which entail the former.

The Best System Analysis

By far the most influential Humean analysis is that of Lewis (inspired by Ramsey and Mill). It agrees that simplicity is key, but implements that in a more flexible way:

The Best System Analysis

To be nomically necessary is to be a consequence of the set of truths about the world that does the best overall job of combining simplicity and strength (the 'best system').

- Giving a rigorous quantitative definition of simplicity and strength, or the balance of the two, seems impossible—it's inevitably going to be pretty vague. But it seems reasonable to think that the different resolutions of vagueness will agree as regards what's *actually* nomically necessary.

The Best System analysis of chance

There's also an extension of the BSA to chance, which Lewis worked out a bit later (1995). Here it is, in streamlined/simplified form:

The Best System Analysis of Chance

For the objective chance of p to be x is for p to be mapped to x by the function that satisfies the probability laws and does the best overall job of combining simplicity and 'fit'.

where 'fit' is a matter of 'assigning high probability to the actual course of events'.

Objections to the Best System Analysis

Putative counterexamples

According to Carroll, there are two distinct possible worlds that satisfy the following descriptions:

w_1 : All particles are X-particles; there are Y-fields; no particle ever enters a Y-field; it's a law that all X- particles in Y-fields have "spin up"; there are no laws other than this and its consequences.

w_2 : The pattern of 'particular matters of fact' is exactly the same as in w_1 . But it's not a law that all X- particles in Y-fields have "spin up"—it's a law that all X-particles in Y-fields have "spin down", and the only laws are this and its consequences.

- Despite what Carroll says, Lewis will definitely **not** agree that these are both possibilities.

Does lawhood explain truth?

Suppose it's a law that p . Most philosophers agree that this entails that it's true that p .

But does the fact that it is a law that p **explain** the fact that p ?

- ▶ According to most anti-Humeans the answer is straightforwardly yes. But it seems hard for Humeans to agree.

On the Lawlike Truth theory, its being a law that p is just p 's being true and lawlike.

But it seems odd to think that a conjunction can explain one of its conjuncts—it seems the other way around!

Something similar seems to hold on the BSA. It's partly because p is true that p belongs to the set of truths that best combines simplicity and strength.

Do laws govern?

Suggestion: we can understand the question whether laws “govern” what happens as equivalent to the question whether lawhood explains truth.

Beebe thinks that although this is something favoured by ‘unreflective common sense’, Humeans should just reject it.

She also suggests that the intuition that worlds like Carroll’s w_1 and w_2 are possible depends on this ‘governing’ conception, and has no force if we give up on governing.

A giant coincidence?

Anti-Humeans often voice a complaint along the following lines:

‘According to Humeans, the fact that [nothing ever travels faster than light] has no explanation at all. So if Humeanism is true, this fact is simply a gigantic coincidence. And this is an absurd thing to believe!’

The idea is that unless we thought that there might be some explanation of the fact that all Fs are Gs, it would not be rational to come to *believe* that all Fs are Gs just on the basis of, e.g., examining lots of F things and finding that they are Gs.

A giant coincidence?

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What, according to **anti**-Humeans, explains the fact that it's a *law* that nothing goes faster than light?

- ▶ If the answer is 'nothing': why is it better to stop there than with the fact that nothing goes faster than light?
- ▶ If the answer is 'the fact that it's a law that it's a law that nothing travels faster than light': isn't this just as pointless and unsatisfying as 'turtles all the way down'?

The modal logic of nomic modality

A very annoying clash of terminology

Carroll defines 'physical' modality as follows:

[A] proposition is physically necessary if and only if it is true at all possible worlds with all and only the laws of the actual world.

I.e. ' p is physically possible' will mean ' p is consistent with the truth about which propositions are and are not laws'. To avoid ambiguity, call this "same-laws" possibility.

Some principles of modal logic

Consider the following two principles about nomic necessity:

4 (Iteration)

It is metaphysically necessary that for all p : if it is nomically necessary that p , then it is nomically necessary that it is nomically necessary that p .

B (Brouwersche Axiom)

It is metaphysically necessary that for all p : if p , it is nomically necessary that it is nomically possible that p

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Both are logically guaranteed for same-laws possibility. But for nomic possibility understood in the usual way, many Humeans will reject them.

Some principles of modal logic

The Lawlike Truth theory upholds 4: since lawlikeness is necessary, the collection of all lawlike truths necessitates that all of its members are lawlike truths.

But it does not uphold B. There could be a lawlike falsehood that is compatible with all the laws:

E.g.: maybe *there are no particles* is false but lawlike and nomically possible: if it had been true, it would have been nomically necessary.

The BSA and the modal logic of nomic necessity

The same problem for B arises on the BSA.

We also get failures of 4. Consider a world where Newton's laws hold without exception except that there is a certain very specific arrangement of 17 particles such that whenever 17 particles are arranged that way, the one in the middle accelerates without the others accelerating at all. This happens 94 times in the history of the universe.

Plausibly at such a universe a generalization saying what happens when we have the special arrangement will be a law. But it seems that at such a universe, it might be nomically possible that the special arrangement never happens at all and Newton's laws hold without exception.

And if it never happened, Newton's laws would win out over the weird laws, so it would no longer be nomically necessary that the acceleration happens whenever we have the special arrangement.